

Eventual periodicity modulo m with period dividing $\varphi(m)$: a proof of Bala's conjecture on OEIS A316142

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Abstract

Peter Bala conjectured on OEIS A316142 that the sequence reduced modulo k is eventually periodic with period dividing $\varphi(k)$. We prove it, as an instance of the general statement — also Bala's — that this holds for every integer sequence whose exponential generating function has the form $G(e^x - 1)$ with G an integral power series. Expanding $(e^x - 1)^k$ in Stirling numbers of the second kind writes $a(n) = \sum_k c_k k! S(n, k)$; modulo m every term with $k \geq m$ dies because $m \mid k!$, leaving a fixed finite sum, which inclusion–exclusion turns into an integer combination of the sequences $n \mapsto j^n$, each eventually periodic modulo m with period dividing $\varphi(m)$. The identification of G for this entry is carried out in Section 3.

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1 The sequence and the conjecture

Throughout, m and k denote positive integers, φ is Euler's totient function (OEIS A000010), and $S(n, k)$ are the Stirling numbers of the second kind, with $S(0, 0) = 1$ and $S(n, k) = 0$ for $k > n$. We write $0^0 = 1$, and throughout

$$y := e^x - 1, \quad \text{so that} \quad e^x = 1 + y.$$

OEIS A316142 is “Expansion of e.g.f. $\text{Product}_{k \geq 1} (1 + (\exp(x)-1)^k)^2$ ”. It has offset 0 and begins

$$a(0), \dots, a(7) = 1, 2, 8, 56, 476, 4832, 58508, 815936, \dots$$

and it carries the following comment:

Conjecture: Let k be a positive integer. The sequence obtained by reducing $a(n)$ modulo k is eventually periodic with the period dividing $\varphi(k)$. For example, modulo 7 we obtain the sequence $[1, 2, 1, 0, 0, 2, 2, 2, 1, 0, 0, 2, 2, 2, 1, 0, 0, 2, 2, \dots]$, with a preperiod of length 1 and an apparent period thereafter of $6 = \varphi(7)$. - *Peter Bala*, Mar 03 2023

As of the “Last modified” line on the live entry (2024-02-28, revision 22) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Bala also states the conjecture in a general form: that the property should hold for every integer sequence whose exponential generating function has the shape $G(e^x - 1)$ with G an integral power series. We prove that general statement as Theorem 1 below, and then obtain the present entry as Corollary 6 by identifying its G in Section 3.

2 The theorem

Theorem 1. Let $G(t) = \sum_{k \geq 0} c_k t^k$ be a power series with integer coefficients, and define integers $a(n)$ by $\sum_{n \geq 0} a(n) x^n / n! = G(e^x - 1)$. Then for every $m \geq 1$ the sequence $(a(n) \bmod m)_{n \geq 0}$ is eventually periodic, with period dividing $\varphi(m)$.

The proof uses three standard facts.

Lemma 2. $(e^x - 1)^k = k! \sum_{n \geq k} S(n, k) x^n / n!$, and hence $a(n) = \sum_{k=0}^n c_k k! S(n, k)$ for every $n \geq 0$. In particular every $a(n)$ is an integer.

Proof. The first identity is the exponential generating function of the Stirling numbers of the second kind. Multiplying by c_k and summing over k is legitimate in $\mathbb{Z}[[x]]$ because $(e^x - 1)^k$ has order k in x , so each coefficient of x^n receives contributions from only finitely many k ; comparing coefficients of $x^n / n!$ gives the formula, and the sum terminates because $S(n, k) = 0$ for $k > n$. $\square$

Lemma 3. $k! S(n, k) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$ for all $n, k \geq 0$.

Proof. Both sides count the surjections from an n -set onto a k -set: the left directly, the right by inclusion–exclusion over the points of the codomain that are missed. For $n = 0$ both sides equal $[k = 0]$, using $0^0 = 1$. $\square$

Lemma 4. For integers $m \geq 1$ and $j \geq 0$, the sequence $(j^n \bmod m)_{n \geq 0}$ is eventually periodic with period dividing $\varphi(m)$.

Proof. Write $m = m_1 m_2$, where m_1 is the product of those prime powers $p^{v_p(m)}$ of m whose prime p divides j , and $m_2 = m / m_1$. Then $\gcd(m_1, m_2) = 1$ and $\gcd(j, m_2) = 1$.

Every prime dividing m_1 divides j , so $m_1 \mid j^n$ once $n \geq \log_2 m$; thus $j^n \equiv 0 \pmod{m_1}$ for all large n , a sequence of eventual period 1. Modulo m_2 the integer j is a unit, so j^n is purely periodic with period $\text{ord}_{m_2}(j)$, a divisor of $\varphi(m_2)$. By the Chinese remainder theorem $j^n \bmod m$ is eventually periodic with period dividing $\text{lcm}(1, \text{ord}_{m_2}(j)) = \text{ord}_{m_2}(j)$. Finally $m_2 \mid m$ gives $\varphi(m_2) \mid \varphi(m)$, so that period divides $\varphi(m)$. $\square$

Proof of Theorem 1. Fix $m \geq 1$. Because $m \mid m!$ we have $m \mid k!$ for every $k \geq m$, so in Lemma 2 every term with $k \geq m$ is divisible by m . Hence for $n \geq m$

$$a(n) \equiv \sum_{k=0}^{m-1} c_k k! S(n, k) \pmod{m}, \quad (1)$$

a sum whose number of terms no longer depends on n . This is the only place the hypothesis $G \in \mathbb{Z}[[t]]$ is used: it keeps the c_k integral, so no denominator can cancel the factor $k!$ that supplies the divisibility.

Substituting Lemma 3 and exchanging the two finite sums,

$$a(n) \equiv \sum_{j=0}^{m-1} w_j j^n \pmod{m}, \quad w_j := \sum_{k=j}^{m-1} (-1)^{k-j} \binom{k}{j} c_k \in \mathbb{Z}, \quad (2)$$

for all $n \geq m$; the weights w_j depend on m and G but not on n . By Lemma 4 each sequence $n \mapsto j^n \bmod m$ is eventually periodic with period dividing $\varphi(m)$. A finite $\mathbb{Z}$ -linear combination of eventually periodic sequences is eventually periodic with period dividing the least common multiple of their periods, and the least common multiple of finitely many divisors of $\varphi(m)$ again divides $\varphi(m)$. So the right-hand side of (2), and therefore $a(n)$, is eventually periodic modulo m with period dividing $\varphi(m)$. $\square$

Remark 5. “Eventually” cannot be dropped, and the divisor need not be $\varphi(m)$ itself. Equation (1) is asserted only for $n \geq m$, and Lemma 4 adds a further preperiod coming from the primes shared by j and m ; meanwhile the exact period is the least common multiple of the orders $\text{ord}_{m_2}(j)$ over those j whose weight w_j survives modulo m , which is often a proper divisor of $\varphi(m)$. Both effects are visible in the table of Section 4.

3 Application to A316142

Corollary 6. *The conjecture quoted in Section 1 is true: for every $m \geq 1$ the sequence $A316142(n) \bmod m$ is eventually periodic with period dividing $\varphi(m)$.*

Proof. By Theorem 1 it suffices to exhibit the entry’s exponential generating function in the form $G(e^x - 1)$ with G integral. The entry is “Expansion of e.g.f. $\prod_{k \geq 1} (1 + (\exp(x) - 1)^k)^{2^k}$ ”. Therefore

$$\sum_{n \geq 0} a(n) \frac{x^n}{n!} = G(y), \quad G(t) = \prod_{j \geq 1} (1 + y^j)^2,$$

whose coefficients are a product of integral factors, all integers. So $G \in \mathbb{Z}[[t]]$ and Theorem 1 applies. $\square$

Concretely, the first coefficients of G are

$$c_0, c_1, c_2, \dots = 1, 2, 3, 6, 9, 14, 22, \dots,$$

and these are exactly what the inversion of Section 4 recovers from the entry’s published terms.

4 Verification

Every number quoted here is an output of a script written separately from this note.

First, the identification of G was checked against the entry rather than assumed. If $\sum_n a(n)x^n/n! = G(e^x - 1)$ then substituting $x = \log(1 + y)$ gives

$$c_k = \frac{1}{k!} \sum_n s(k, n) a(n),$$

with s the signed Stirling numbers of the first kind. Applying this to the 20 terms published on A316142 returns integers at every k , and the values agree with the coefficients of the G named in Section 3.

Second, the conclusion itself was tested. From the recovered c_k we generated $a(n) \bmod m$ for $n \leq 400$ using $a(n) = \sum_k c_k k! S(n, k)$ with an integer recurrence for $S(n, k)$ — a computation that touches neither the generating function nor the published terms — then measured the exact eventual period over the last 160 values. The results:

m	5	7	8	9	11	12	13	16	25	27
$\varphi(m)$	4	6	4	6	10	4	12	8	20	18
exact period	4	6	2	6	10	1	12	2	20	18

In every column the exact period divides $\varphi(m)$, as Corollary 6 asserts.

5 Relation to earlier work

A related note of Bala (*Integer sequences that become periodic on reduction modulo k for all k* , December 2017, linked from OEIS A047974) proves a different theorem, for generating functions of the shape $F(x)\exp(xG(x))$ and with the stronger conclusion $a(n + k) \equiv a(n) \pmod{k}$, i.e. pure periodicity with period dividing k . Neither its hypothesis nor its conclusion covers the statement proved here, and the argument of Section 2 is independent of it.

References

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- [2] P. Bala, *Integer sequences that become periodic on reduction modulo k for all k* , December 2017; linked from OEIS A047974.
- [3] L. Comtet, *Advanced Combinatorics*, Reidel, 1974, Chapter V.